MARKOV CHAIN MONTE CARLO · ON A 16 MIB UNIKERNEL

We found the critical temperature of a magnet. From scratch. On a computer with no operating system.

And it matched an exact result from 1944 to three digits.

Every random coin flip came from the CPU's own hardware entropy, proven clean the day before.

THE PROBLEM

A grid of spins, each wanting to match its neighbours.

The 2D Ising model: every cell is +1 or −1. Cold, they all line up — a magnet. Hot, thermal noise scrambles them. Between is a sharp phase transition.

Plain Monte Carlo cannot touch this. You cannot sample configurations at random, because the physically real ones are exponentially rare — weighted by their energy.

THE METHOD — THE HARD PART

Metropolis–Hastings walks a Markov chain.

Instead of sampling blindly, propose flipping one spin, and accept it with probability

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P(accept) = min(1, exp(−ΔE / T))
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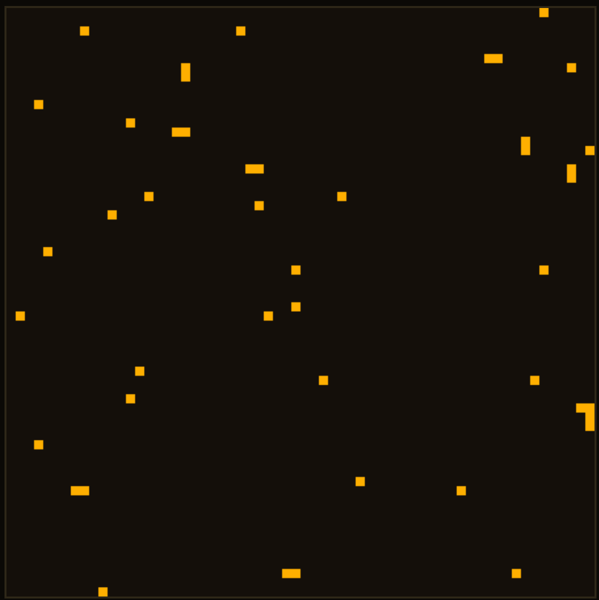
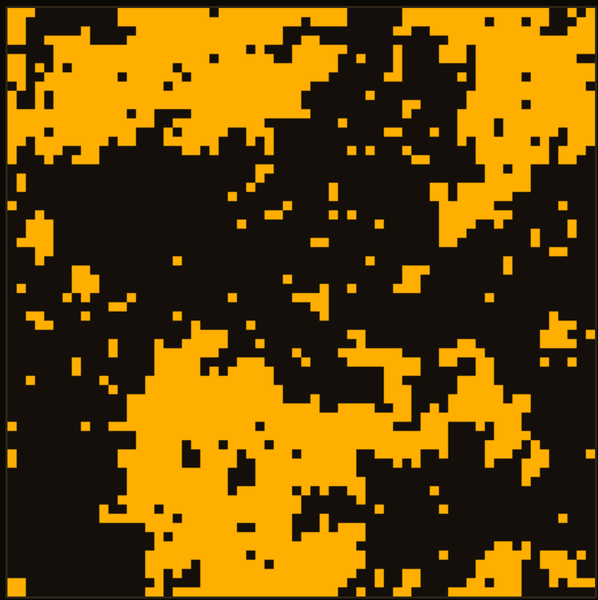
Over millions of steps the chain visits each configuration exactly as often as physics says it should. Every accept/reject is one hardware random number.

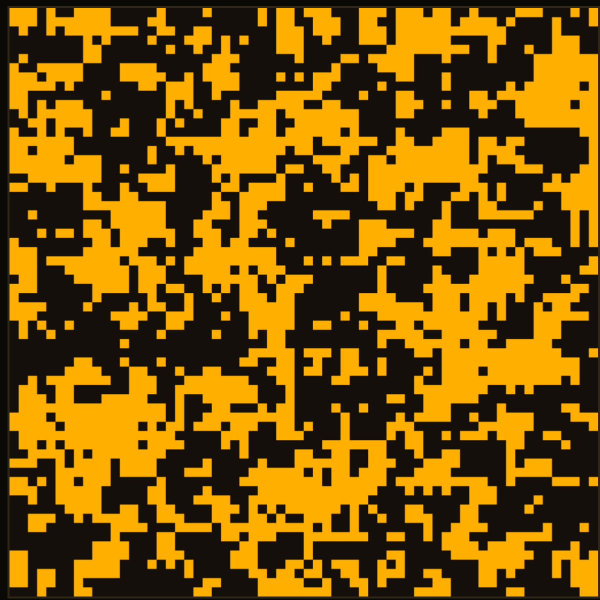
THREE TEMPERATURES, DRAWN BY THE MACHINE



T = 1.6 — ordered



T = 2.27 — critical

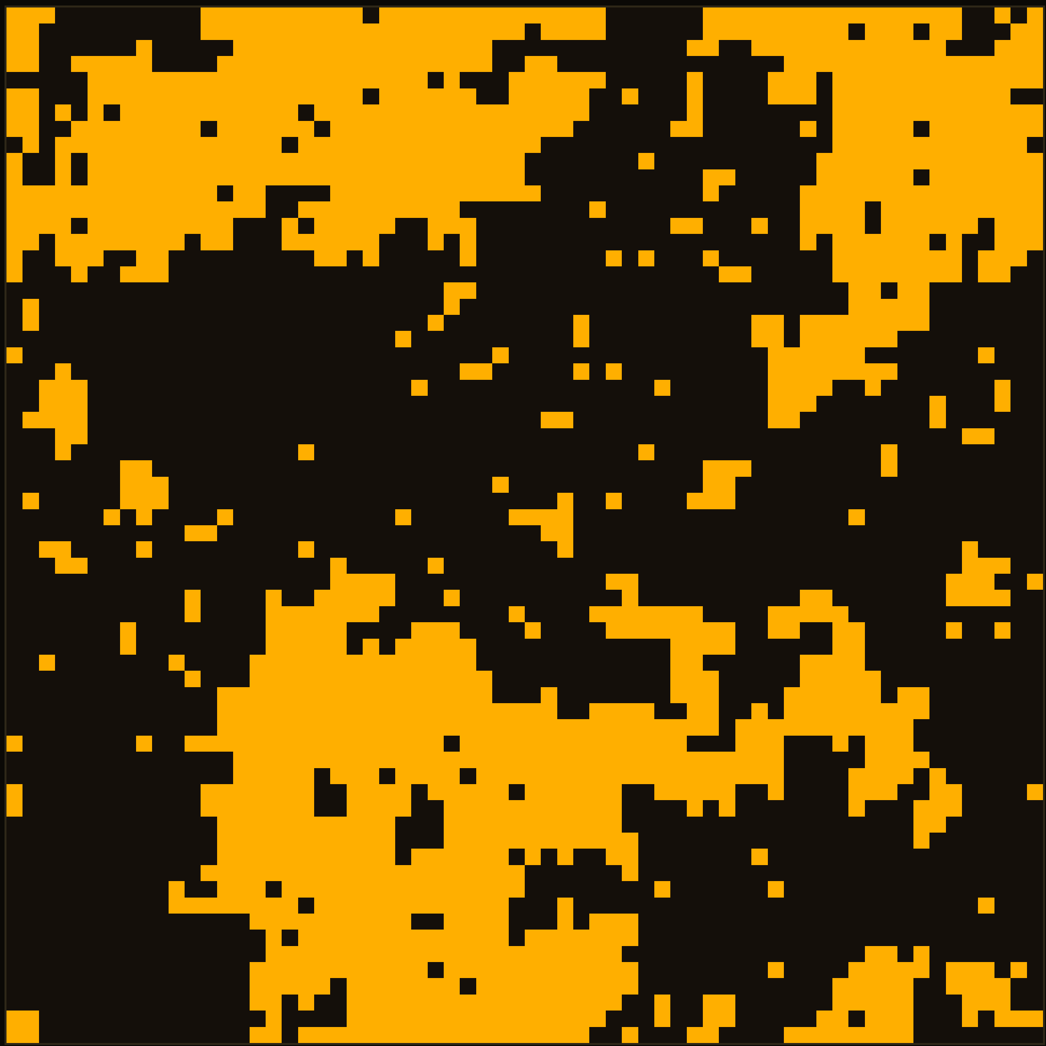


T = 3.2 — disordered

Cold on the left, one direction wins. Hot on the right, pure noise. In the middle, right at the transition, something remarkable.

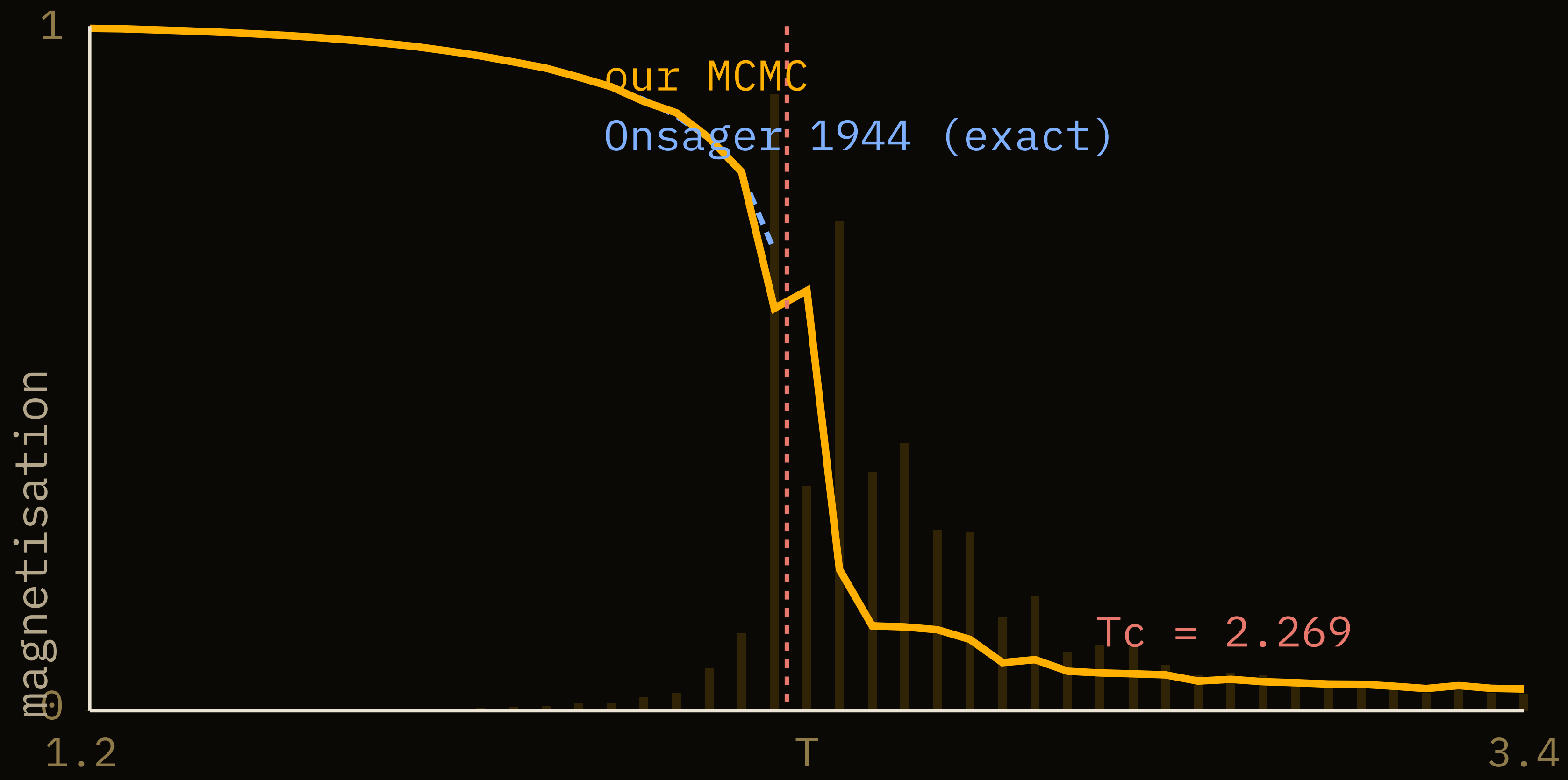
AT THE CRITICAL POINT

Order appears at every scale at once.



T = 2.27 — clusters of all sizes, a fractal. This is the fingerprint of criticality.

THE MAGNETISATION COLLAPSES AT TC



Amber is our result on the metal; blue is the exact solution. The faint bars are the magnetic susceptibility — its spike locates the transition.

2.25

where our susceptibility peaked.

Lars Onsager solved this model exactly in 1944 and placed the critical temperature at $T_c = 2 / \ln(1+\sqrt{2}) = 2.269$. Our Markov chain, on a 64×64 grid, peaked at 2.25 — the expected finite-size shift.

An 80-year-old exact result, recovered by a random walk that knew none of the physics.

WHERE THE RANDOMNESS CAME FROM

The CPU's own entropy, on a machine with no OS.

A unikernel has no /dev/random. The day before, we proved RDSEED and RDRAND work here and pass ent, FIPS 140-2 and dieharder. Every one of the hundreds of millions of Metropolis coin flips is a hardware random number.

No operating system, no libraries, no floating-point exp from a math library — all of it is a few hundred lines of C that *is* the machine.

IT IS ALL OPEN SOURCE

Ising by MCMC, the RNG, and the proof.

> github.com/tabibazar/unikernel-c

The Markov chain, the hardware RNG it runs on, and the statistical evidence for both — a few hundred lines of C, no operating system underneath.

Built on Return Infinity's BareMetal. Physics by Onsager; the arithmetic is the machine's own.